
CSCI 3388 Practice Midterm Exam**Instructions:**

- **DO NOT DETACH ANY PAGES—EVEN THE SCRATCH PAGES—FROM THIS EXAM!** Removing page(s) may result in a deduction in your grade.
- This exam is closed book and closed notebook, with no electronic devices allowed. The only resource you may use is one 8.5×11 -inch double-sided study sheet that you prepared yourself.
- You are not allowed to discuss the exam with anyone until the solutions are published.

Any deviation from these rules will constitute a violation of the University policies on academic integrity.

- The exam is designed for 75 minutes and consists of multiple-choice and written-answer problems.
- For the multiple-choice problems, mark your answers by filling in the bubble to the left of your choice. Select only one choice.
- For the written-answer problems, write your answers clearly in the provided space.
- The exam has 6 pages printed on both sides, including this page and three scratch pages at the end.

Name: _____

Email: _____

Eagle ID: _____

Section	Points
1-4 (Multiple Choice)	/ 20
5-8 (Written Answer)	/ 80
Total	/ 100

Multiple Choice (5 points each)

1. N-gram Language Models

Given the following bigram counts from a corpus:

- $\text{count}(\text{"the"}, \text{"cat"}) = 15$
- $\text{count}(\text{"the"}) = 100$
- $\text{count}(\text{"cat"}) = 20$

What is $P(\text{"cat"} \mid \text{"the"})$ using maximum likelihood estimation?

- ☐ 0.15
- ☐ 0.20
- ☐ 0.75
- ☐ 0.80

2. Neural Network Activation Functions

Which activation function is most commonly used in modern deep learning to address the vanishing gradient problem?

- ☐ Sigmoid
- ☐ Tanh
- ☐ ReLU
- ☐ Linear

3. Attention Mechanism

In the scaled dot-product attention formula $\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d_k})\mathbf{V}$, what is the purpose of the scaling factor $\sqrt{d_k}$?

- ☐ To normalize the attention weights
- ☐ To prevent the dot products from becoming too large
- ☐ To ensure the softmax outputs sum to 1
- ☐ To make the computation more efficient

4. GPT Evolution

Which GPT model first demonstrated significant zero-shot capabilities without task-specific fine-tuning?

- ☐ GPT-1
- ☐ GPT-2
- ☐ GPT-3
- ☐ ChatGPT

Written Answer (80 points total)

5. Attention Scaling (20 points)

- a) Derive why variance of dot product $q \cdot k$ grows with d_k if $q, k \sim \mathcal{N}(0, 1)$.
- b) Show dividing by $\sqrt{d_k}$ stabilizes variance.
- c) What happens to attention weights when d_k becomes very large without scaling? Explain the practical implications for training transformers.

6. Transformer Architecture and Multi-Head Attention (20 points)

You're implementing a transformer model and need to understand the attention mechanism in detail.

- a) (7 pts) Explain how multi-head attention works. What are the benefits of having multiple attention heads?
- b) (7 pts) Given query Q, key K, and value V matrices of shape (seq_len, d_model), derive the computational complexity of self-attention.
- c) (6 pts) How would you modify the attention mechanism to handle very long sequences (10k+ tokens) efficiently?

7. Value Iteration in an MDP (20 points)

Consider a 3-state MDP: (s_1, s_2, s_3) with deterministic transitions and rewards:

- s_1 : A $\rightarrow s_1$, reward 1; B $\rightarrow s_2$, reward 0.
- s_2 : A $\rightarrow s_2$, reward 2; B $\rightarrow s_3$, reward 0.
- s_3 : any action $\rightarrow s_3$, reward 5.

Discount $\gamma = 0.9$.

- a) (8 pts) Write the Bellman optimality equation.
- b) (8 pts) Starting with $V_0(s) = 0$, perform one value iteration update. Compute $V_1(s_1), V_1(s_2), V_1(s_3)$.
- c) (4 pts) Which action is preferred in s_1 after one iteration?

8. GPT Alignment & Safety (20 points)

Your LLM is generating harmful content and you need to implement RLHF to fix this.

- a) (10 pts) How would you design the human preference data collection process for RLHF?
- b) (10 pts) Your model is reward hacking (optimizing for the reward model instead of being helpful). How would you detect and fix this?

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